

Following Evidence About the Same Part: An Instance-Aligned Benchmark on LDraw Assemblies

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Abstract

A correct answer about a part does not establish that a vision-language model followed the evidence identifying that part. The answer may follow a prior, persist after relevant evidence changes, or change under an irrelevant display perturbation. We introduce BrickAtlas, an instance-aligned measurement protocol on 24 human-authored LDraw assemblies with 15,334 placed parts. Its main question is whether answers about the same target correctly follow changed visual evidence and remain correct under specified semantic invariances. The 617-item resource separates 140 visual questions, 146 graph questions and two classes of controls. Paired observations bind canonical identities, exact inputs, condition-specific gold answers and answerability decisions. New-evidence accuracy and paired both-correct distinguish adaptation from disruption. An audit of the unchanged graph questions illustrates the need: counting nodes alone attains 86.30% accuracy, but zero both-correct on answer-changing graph pairs, while a graph algorithm solves both sides. A separate matched-graph extension controls additional surface statistics. We report verified corpus and algorithmic evidence; visual model comparisons and human answerability review remain unrun. The contribution is a reproducible resource and falsifiable protocol, not a completed model ranking.

1. Introduction

In a dense assembly, two pieces can share geometry while differing in color, position or identity. A question about one piece therefore asks more than whether a familiar shape is present: an answer must remain bound to the designated instance. If the visible assignment of labels to pieces changes, a model should follow the new evidence. If only the names used to identify pieces change consistently, the semantic answer should remain correct. Ordinary accuracy observes neither distinction.

This motivates our primary research question: *for the*

same identified part in an original assembly, do answers correctly follow task-relevant visual changes and remain correct under specified answer-preserving changes? We study Color and Part-type separately. Two supporting questions concern the total effect of hiding non-operands at a fixed camera and whether graph answers follow relations beyond elementary input statistics.

Human-authored LDraw models [4] provide useful test conditions: repeated instances, densely nested assemblies, author step records and geometry at preserved source poses. Instance identity links these assets without requiring them all in one question. Figure 1 illustrates the distinction. A visual question uses a numbered image; a separate graph question uses an explicit edge list; a visibility edit uses declared actions. Shared identity permits controlled inspection, but does not imply a within-item multimodal reasoning chain.

Related resources already supply important ingredients. BrickNet [3] has instance and connector graphs; LEGO-Puzzles [6] uses editable scenes and step-consistent renders; PhyBlock [5] combines scene identity, dependencies and simulated assembly. Our distinction is the explicitly paired measurement: an identified target, exact observations, two semantic golds, and a decision on whether each observation makes the question answerable. A change in old-gold accuracy alone cannot establish that an answer followed new evidence.

The present empirical evidence is algorithmic. On all 73 natural deletion questions, a predictor that ignores edges and returns the node count minus one answers 63 correctly. On the corresponding fixed-node, answer-changing pairs it answers neither pair jointly correctly; connected-component algorithms solve both. This post-hoc replication exposes a concrete interpretive gap between high accuracy and evidence following. It is not a result about a learned model.

Our contributions are threefold. First, we make same-instance visual evidence following measurable on unchanged source assemblies, with modular structural and execution controls. Second, we define paired semantic measures and source-balanced estimates that separate correct

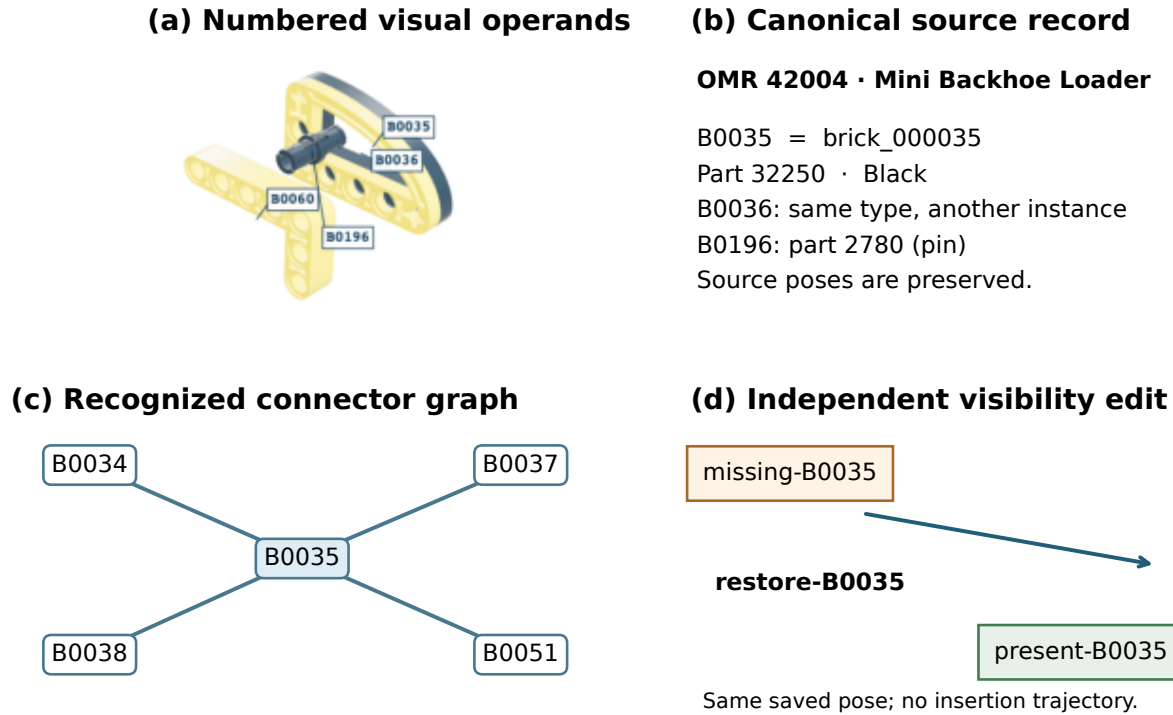


Figure 1. One source instance, separately scored contracts. In OMR 42004, B0035 is a black part 32250 and B0036 is another instance of the same type. (a) Detail from the numbered operand image; other source parts are hidden, with remaining poses unchanged. (b) Canonical source records disambiguate repeated types. (c) Recognized axle edges link B0035 to four other instances; this schematic does not depict spatial placement. (d) An independent action restores visibility at the saved pose. Records and graph facts are not supplied to the visual question. The links support inspection across tasks, not a single cross-module inference chain.

adaptation, stable error and information withdrawal. Third, we supply an audited natural distribution and an independently verified hypothetical graph challenge. The full question index, input provenance, baselines and executable analysis accompany the resource. The visual research question remains open until exact-input human review and genuine model responses are available.

2. Related Protocols

Compositional VQA resources such as CLEVR [2] and GQA [1] motivate specifying what operations and evidence a question requires. Our focus is narrower: following evidence about an identified part under paired observations, with separate controls for source lookups, graph operations and display actions. Table 1 compares the three closest assembly protocols. The supplement records paper pages, exact versions and inspected official code symbols.

BrickNet generates graph-backed programs and measures parse, connectivity, collision and perceptual properties. Distinct placed instances and connectors are already explicit in its representation. LEGO-Puzzles evaluates spatial and sequential VQA, sometimes jointly using current, target and part images. Its v3 protocol includes

three-annotator verification and editable rendered scenes. PhyBlock evaluates planning against legal dependency orders and physical understanding through static and dynamic questions. Its scene records identify target instances, while candidate IDs in planning can denote reusable block variants. These are substantial capabilities, not absences that our use of IDs resolves.

The same-target, new-gold and both-correct VQA estimands used here are not reported in these pinned versions. Their assets could support extensions with this protocol. Adding IDs and templates can establish which source entity a question concerns; it does not by itself specify how the answer should change for a second rendered observation, whether that observation is decidable, or which semantic transition a raw response exhibits. Those additional records define the object of measurement in BrickAtlas.

3. Benchmark and Observation Contracts

3.1. Source assemblies and identity

The resource contains 24 original assemblies, 15,334 placed instances and 617 questions. Source files, part transforms and original gold answers are preserved. Each canonical

Table 1. Related protocols and the evidence they measure. Comparisons use BrickNet’s CVPR 2026 paper, LEGO-Puzzles arXiv v3 and PhyBlock arXiv v2. “Not reported” refers to the specific paired VQA estimand, not an inability to implement it. Full page- and code-level evidence is in the supplement.

Protocol	Instance alignment and observations	Controlled comparison	Measured outcome
BrickNet [3]	Graph-local part and connector IDs; generates programs from a start token or caption.	Graph/pose representations, sampling and training ablations.	Parse/connectivity and collision validity; render-text agreement. Our paired VQA estimand is not reported.
LEGO-Puzzles [6]	Image tokens, option labels and arrows; multiple current/target/part images support spatial and sequential questions.	Fixed camera across steps; editable scene attributes; step length and prompt comparisons.	VQA/ordering accuracy and human-rated image generation. Our paired VQA estimand is not reported.
PhyBlock [5]	Scene-instance IDs and dependencies; reusable candidate IDs in planning; target/candidate images and optional feedback.	Pose-constrained/topology-oriented scoring, stepwise feedback and simulated perturbations.	AOV step precision/recall and physical VQA accuracy. Our paired VQA estimand is not reported.
BrickAtlas	Canonical instance linked to images, records, graphs and display edits; each item has one declared contract.	Same-target fact change, semantic invariance and evidence withdrawal, with separately bound golds.	New-evidence accuracy and paired both-correct; source-cluster contrasts. Learned-model results remain unrun.

identity combines a source hash with an instance ID. A display label such as B0035 is unique within its source, not across the corpus. Repeated part numbers remain different instances. Source-size bands D1–D4 describe corpus scale; they are not validated cognitive difficulty levels.

The images use actual LDraw geometry. Hiding non-operands and replaying visibility changes do not relocate any part. Such partial observations can omit supporting pieces; they represent access to an unchanged assembly, not a proposed physical arrangement. Recognized connector relations establish a limited graph, not exhaustive contact, gravity stability or mechanical feasibility. Unresolved physical candidates are retained for review.

3.2. Task roles and scope

Every item has one declared model-visible contract: a numbered image (V), selected source records (S), explicit graph evidence (G), or facts and discrete display actions (E). The scorer’s source records, references and alternate gold are separate from the request. No family is presented as a full perception-to-assembly pipeline.

The four reporting roles are disjoint. Core visual questions comprise 73 Color and 67 Part-type items. Structural evidence comprises Neighbors and Graph deletion, 73 each. The 290 nonconstant controls test coordinate distance, interface records, coverage flags, author-step lookup and visibility restoration. The 41 constant controls contain 24 evidence-limit responses and 17 fixed step programs. All items remain in the complete report; no composite capability ranking mixes these roles.

Color selects the rendered target’s color category. Part-type asks which numbered candidate shares its source type while ignoring color and pose. Near-identical geometry can

make a source-derived type answer visually undecidable; exact-input QA must resolve that boundary. Neighbors returns an exact set from the supplied local graph. Graph deletion returns the number of connected components after removing the target, counting isolated survivors and deduplicating undirected edges.

Execution controls expose preconditions, forbidden facts, effects, budget and goals. A legal restoration makes one hidden instance visible at its saved pose. The given-order replay in Figure 2 follows six declared actions. Invalid actions do not change the state. These controls measure contract execution and output compliance, not mechanical planning.

3.3. Paired observations

Let a pair be $(i, s, o_A, o_B, y_A, y_B, \tau)$, where i is the parent question, s the source, o the exact observation, y its semantic gold and τ its truth policy. We distinguish three policies. *Fact change* assigns two distinct observation-conditioned answers. *Semantic invariance* retains the canonical answer after mapping display labels back to identities. *Information withdrawal* retains the original gold for a dependency comparison while removing evidence needed to infer it.

The primary fact-change comparison pairs operand-only with wrong-image. For Part-type, a displayed candidate-label swap changes which option identifies the matching geometry. For Color, a donor image provides different visual content; shape can change too, so this is not a color-only causal intervention. Original source-gold scores remain available as auxiliary dependency statistics. The new analysis adds the image-conditioned gold without overwriting those scores.

Invariance comparisons cover consistent ID renaming,

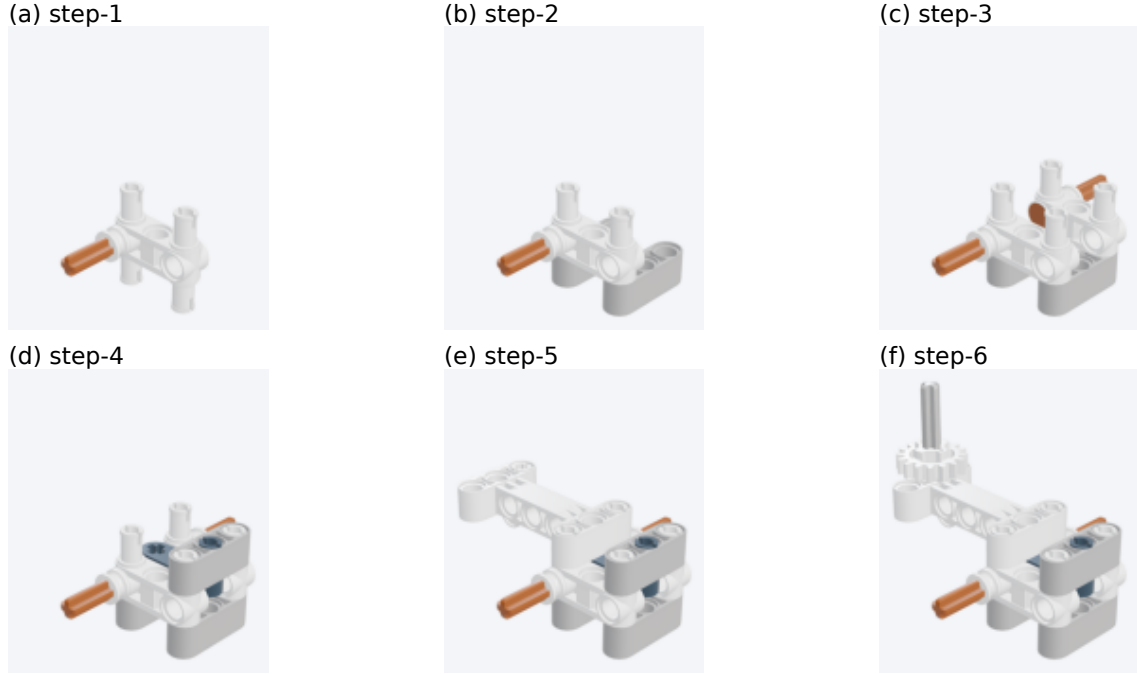


Figure 2. Given-order visibility replay for the supplied author window of set 42102. Panels (a)–(f) show the accepted actions step-1 through step-6, with a common paper crop and original source transforms. The program is supplied by the contract and is constant across this control family. The frames demonstrate action-to-display correspondence; they do not establish discovery of an assembly order, an insertion trajectory or stability.

camera changes and type-preserving color appearance changes. The existing ID permutation uses longer labels and can move text boxes, so it measures stability under naming and display changes rather than a pure identity-only effect. Semantic invariance is necessary but insufficient for visual answerability: a changed view can hide a distinguishing feature.

4. Measurement Validity

4.1. Priors and retained denominators

Answer concentration has different meanings across families (Figure 3). A frequent color category supports a global constant predictor. An answer keyed by a source-specific instance rarely repeats even if the operation is trivial. A small exact-key repetition rate therefore does not certify reasoning difficulty.

Graph deletion illustrates a stronger input-conditioned prior. For target t , let n' be the number of surviving nodes and c' their component count. Define the analysis-only deficit

$$d = n' - c'. \quad (1)$$

If $d = 0$, returning $|V| - 1$ is correct without reading any edge. We retain all questions and report both $d = 0$ and $d > 0$ strata. These strata describe the corpus and never

enter model prompts. Removing the easy stratum would change the target distribution.

4.2. Independent gold and input checks

Node-only prediction accepts only a public node list. A later scoring phase joins those saved predictions with references and graph statistics. Breadth-first search (BFS) and an independently implemented union-find procedure agree on every original and intervened deletion gold. They explicitly handle duplicate edges, isolated nodes and target deletion. Graph oracles validate supplied evidence; they do not certify that all physical connections were recognized.

Each paired record binds source and dataset hashes, exact model revision, adapter, prompt, both request bodies, normalized predictions, gold provenance and truth policy. Validation reconstructs the wire request and parses the saved raw response. It rejects mismatched revisions and duplicate pairs. A missing B response remains a failed observation in the declared denominator; it is not removed by taking the intersection of successful requests.

4.3. Answerability and quality decisions

Visual accuracy is interpretable only for inputs that support the question. Review binds the actual image bytes and dimensions sent in the request, not a higher-resolution source

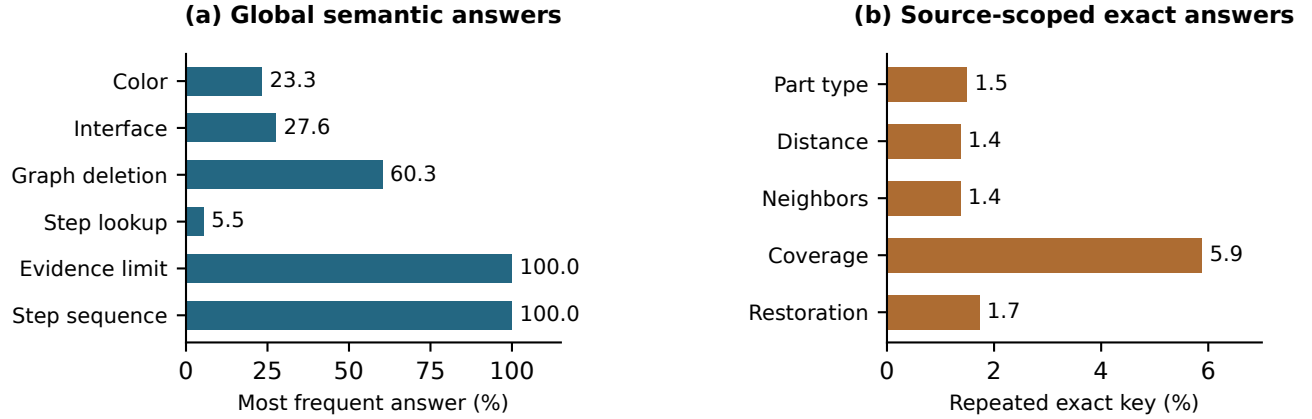


Figure 3. Two different corpus diagnostics. (a) The frequency of the most common global semantic answer exposes category and fixed-output priors. (b) Exact source-scoped answers include the source and instance identity; low repetition is partly a consequence of that key space. The panels use separate scales and are not comparable measures of difficulty. In particular, the global constant-answer prior for graph deletion is weaker than the input-conditioned node-count baseline in Table 4. These are corpus counts, not model scores.

render or an extra view. A paired visual comparison requires independent answerability decisions for both observations. Pending, failed and access-difference cases remain explicitly enumerated alongside the full-denominator result. No model-dependent filtering is permitted.

The current review inventory contains 140 visual items, 111 other flagged items and 573 source-local intersection candidates. A candidate can reflect intended mating, mesh approximation, a source defect or an unresolved case; it is not automatically a collision error. Software validation and the paper’s enlargements do not count as human certification.

5. Evidence-Following Evaluation

5.1. Measures and aggregation

For N fixed pairs with normalized predictions p_A, p_B , the primary fact-change measures are

$$\text{NewAcc} = \frac{1}{N} \sum_i [p_{Bi} = y_{Bi}], \quad (2)$$

$$\text{Both} = \frac{1}{N} \sum_i [p_{Ai} = y_{Ai}] [p_{Bi} = y_{Bi}]. \quad (3)$$

The optional conditional adaptation rate divides the same both-correct numerator by $\sum_i [p_{Ai} = y_{Ai}]$. A zero conditional denominator returns undefined, not zero. We always retain the unconditional rates and display the conditional denominator.

For wrong-image, B outcomes partition into the new answer, the old answer, another legal answer, and invalid or missing output. For invariance, we report both correct, A-

only correct, B-only correct, both wrong with the same semantic answer, and both wrong with different answers. Invalid outputs carry a separate reason and do not become evidence of semantic agreement. Set answers are order-independent, but illegal duplicates are not silently repaired; action answers use the executable contract.

These measures resolve two ambiguities. A model can be correct on both graphs while its accuracy difference is zero. Conversely, an answer can change without either answer being correct. A consistently wrong answer is also distinct from correct retention. Output changes and old-gold accuracy remain useful auxiliary statistics, not substitutes for paired correctness.

For pair-level values z_{spv} , the source-balanced estimate is

$$\hat{\mu} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{|P_s|} \sum_{p \in P_s} \frac{1}{|V_{sp}|} \sum_{v \in V_{sp}} z_{spv}. \quad (4)$$

Thus variants first average within a parent, parents within a source, then sources equally. We also report micro rates, parent counts and source counts. A family absent from a source contributes no zero. Percentile 95% intervals use 10,000 source-cluster bootstrap draws with seed 20260919. Model–baseline differences are computed on the same paired items before resampling. Repeated observations do not increase the number of independent sources.

5.2. Primary question: visual evidence following

The visual experiment asks whether new-image accuracy and both-correct remain high for a fact change, and whether correct retention survives specified invariances. Color and Part-type remain separate (Table 2). Lower accuracy against

Table 2. Core visual reporting, with fixed family denominators. No validated learned-model results are available: “–” means not run, never zero. Human answerability is pending. Family-specific results replace an all-task capability ranking.

Family / model	<i>N</i>	Sources	Source macro (%)
Color	73	24	–
Part-type	67	24	–

the old gold is insufficient: it can represent adaptation, confusion or invalid output. The four B-outcome categories reveal which occurred.

A concrete source example makes the contrast testable. In OMR 42004, B0035 and B0036 share part type 32250, while B0196 is a pin. With fixed options A:B0196 and C:B0036, the operand image’s gold is C. Swapping the visible B0036/B0196 labels makes the image-conditioned gold A. The source CAD remains unchanged. $C \rightarrow A$ is the desired paired response; $C \rightarrow C$ retains the old answer; $C \rightarrow B$ is another legal error. These are scoring examples, not observed model predictions. The supplement includes both actual images and their provenance.

The result index currently contains only unrun learned-model records. Exact requests and alternate gold mappings are validated, but no claim about visual model sensitivity, adaptation or robustness follows from that check. The protocol can falsify evidence following once genuine responses and condition-specific QA exist.

5.3. Supporting question: background and view access

Full-scene versus background-mask holds camera, scale, lighting and label placement fixed (Figure 4). Its primary contrast is the paired source-macro accuracy difference. Hiding non-operands both removes context and may expose surfaces; those effects are not separately identified. Crop changes magnification and cannot isolate background effects. Operand-only versus multi-view measures access to two fixed views, not adaptive exploration.

For each visual comparison, the release preserves the planned set, the jointly decidable set, and reasons for pending or excluded interpretation. Differences are accompanied by source counts and clustered intervals. An interval containing zero does not establish equivalence. No unsegmented screenshot is used to invent an occlusion percentage or projected target area. With no learned-model responses, the direction and magnitude of these access effects remain unknown.

Table 3. Structural family reporting on the natural distribution. “–” means learned-model inference is unrun. Deletion requires the algorithmic controls in Table 4 and paired success; an ordinary family score alone is insufficient.

Family / model	<i>N</i>	Sources	Source macro (%)
Neighbors	73	24	–
Graph deletion	73	24	–

5.4. Supporting question: graph evidence beyond priors

The node-only baseline reaches 63/73 (86.30% micro; 83.75% source macro, 95% interval [72.92, 93.12]). The interval concerns sampling variation across these source clusters; the exact corpus count itself has no sampling uncertainty. Only 10 items from 9 sources have positive deficit. This small stratum limits conclusions about natural surviving-edge difficulty.

The existing intervention adds one edge between surviving components, keeping the node set fixed and reducing the gold component count by one. Table 4 shows zero both-correct for node-only and always-four predictors, versus 100% for BFS/union-find. The contrast establishes that ordinary high accuracy on this distribution can be obtained without consulting the defining relations. It does not establish that any learned model uses this shortcut. The node-prior concern was already known before this revision; the replication and stratification are post-hoc audits.

To control additional surface statistics, a separate extension fixes three observations per parent: a hypothetical anchor, an answer-changing partner, and an isomorphic answer-preserving variant. Its 219 observations form 146 pairs: 37 increases, 36 decreases and 73 unchanged answers. All pairs match node set, unique edge count and target degree. 12 answer-changing pairs also match the surviving degree sequence; the other 61 parents have only four or five survivors, for which the declared templates do not provide that match.

For six or more survivors, a six-cycle versus two triangles has different component counts despite identical per-node degrees. For four or five survivors, a path versus triangle-plus-isolates matches counts but not degrees. Additional survivors are isolated. These graphs deliberately change the evidence distribution and are labeled hypothetical; they are not newly inferred source connections. They add no source and never replace the natural questions.

All extension golds agree under independent graph algorithms. Rules and shuffled opaque observation IDs were fixed before new model responses. Each observation receives a fresh request without the transformation or answer direction. The extension’s algorithmic verification is complete, while learned-model inference is unrun. Natural,

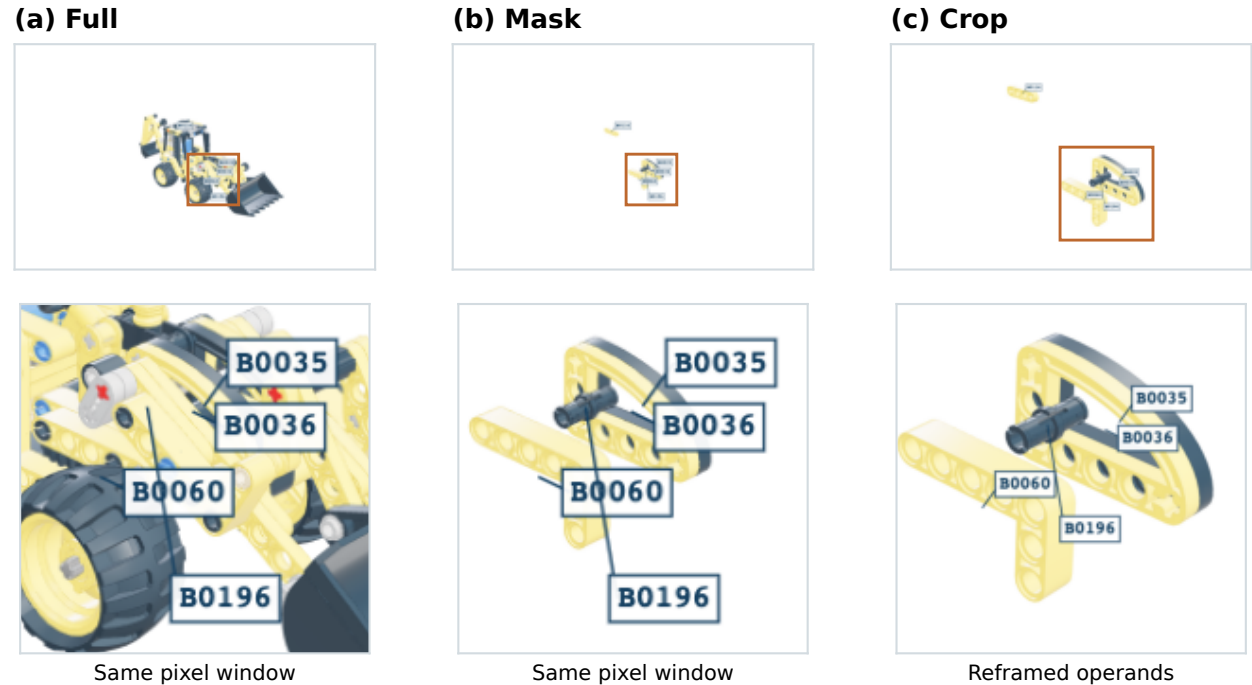


Figure 4. Full / Mask / Crop for the same Part-type item. Top: each original frame with the paper’s detail window. Bottom: enlarged details; Full and Mask use exactly the same pixel rectangle. Full/Mask fix camera, scale, lighting and label placement, while hiding non-operands can reveal occluded surfaces. The comparison measures the total effect of this access change, not pure attention. Crop additionally reframes the operands. All enlargements and orange boxes exist only in the paper and are not model inputs. Original poses are preserved, and answerability remains subject to exact-input QA.

Table 4. Measured algorithmic controls on the unchanged 73 deletion questions and their original/intervened pairs (24 sources). Both is the source-macro rate of answering each pair correctly. Natural source-macro intervals are 95% source-cluster intervals, reported in the text and supplement. These are executable baselines, not learned-model results.

Algorithmic baseline	Micro (%)	Source macro (%)	Both (%)
Node count only	86.30	83.75	0.00
Always four	60.27	61.11	0.00
BFS / union-find	100.00	100.00	100.00

original-intervention and matched-challenge results must remain separately reported.

6. Discussion and Resource Use

This resource isolates a modest but important requirement: answering about an identified part should follow the evidence made available for that question. Preserved source identity supports controlled observations; condition-specific gold and raw semantic responses make correct adaptation distinguishable from disturbance. The algorithmic audit demonstrates the interpretive value of that distinction on the present graph distribution.

Several limits determine the scope of future conclusions. The corpus has only 24 human-authored sources, and shared parts or authoring conventions can create dependence be-

yond a source cluster. Visual answerability is unresolved. Connector coverage is incomplete, display edits do not model forces, and the matched challenge is an artificial graph distribution. Modular task scores cannot locate a model’s internal cognitive failure or demonstrate end-to-end assembly.

The supplement contains all 617 questions and golds for audit, full family/control tables, source coverage, protocol evidence and failure cases. Reference-bearing materials are evaluator resources, not inference inputs. Future training splits must keep every source and all its views and variants together. Reusing public sources also requires explicit contamination and source-rights review.

The analysis separates unavailable evidence from measured zero. API errors, timeouts, refusals and invalid out-

puts remain in the denominator with distinct reasons. A completed result enters the paper only after its manifest, raw receipts, semantic scoring and sample counts pass validation. The current contribution therefore remains a method and resource with measured algorithmic controls. Visual and physical conclusions require the corresponding human and model evidence.

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